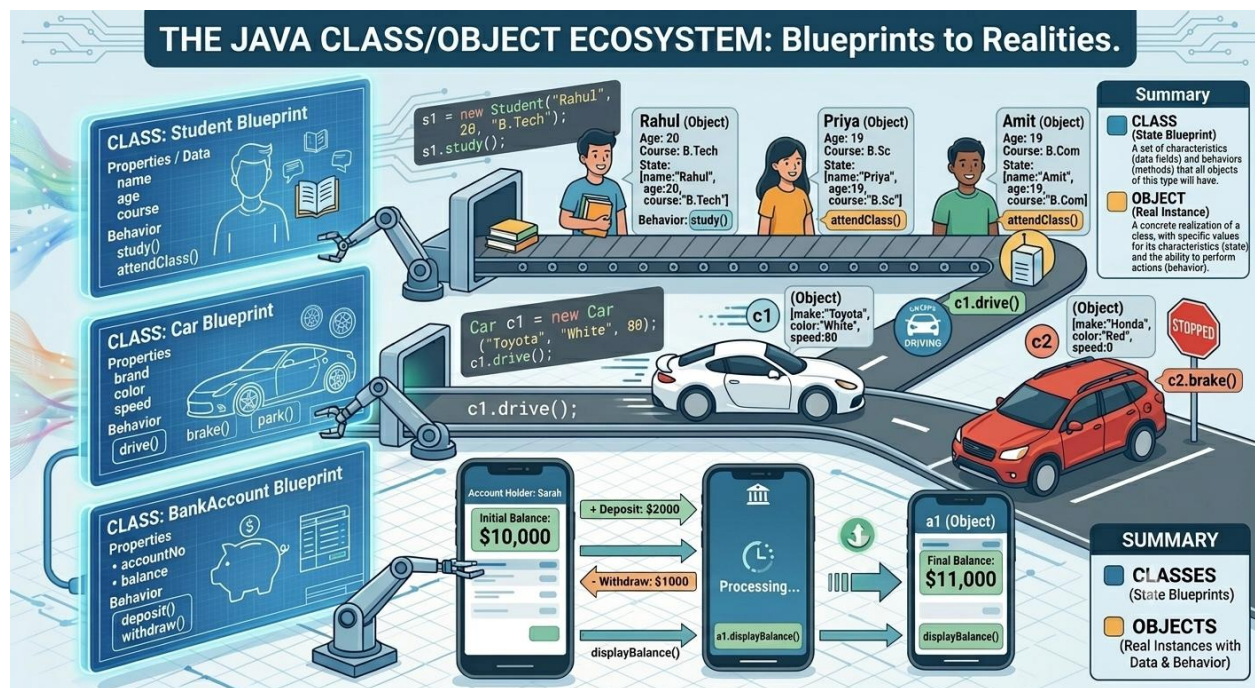


In Java, a **class** can be understood as a **blueprint or template** for creating objects. A real-world entity such as a **Student, Car, Bank Account, Employee, or Mobile Phone** can be represented as a class.

Simple idea

Think of a **class as a design** and an **object as the actual thing**.



For example:

Class: Student

Objects: Rahul, Priya, Amit

The `Student` class describes what every student can have and do.

1. Student Class

```
class Student {  
    // Properties / Data  
    String name;  
    int age;  
    String course;  
  
    // Method / Behaviour  
    void study() {  
        System.out.println(name + " is studying.");  
    }  
}
```

```

    }
}

public class Main {
    public static void main(String[] args) {

        Student s1 = new Student();

        s1.name = "Rahul";
        s1.age = 20;
        s1.course = "B.Tech";

        System.out.println(s1.name);
        System.out.println(s1.age);
        System.out.println(s1.course);

        s1.study();
    }
}

```

Real-world entity: Student

Properties: Name, age, course

Behaviour: Study

2. Car Class

A car is another real-world entity.

```

class Car {
    String brand;
    String color;
    int speed;

    void drive() {
        System.out.println(brand + " is driving.");
    }

    void stop() {
        System.out.println(brand + " has stopped.");
    }
}

public class Main {
    public static void main(String[] args) {

        Car c1 = new Car();

        c1.brand = "Toyota";
        c1.color = "White";
        c1.speed = 80;
    }
}

```

```

        System.out.println(c1.brand);
        System.out.println(c1.color);
        System.out.println(c1.speed);

        c1.drive();
        c1.stop();
    }
}

```

Real-world entity: Car

Properties: Brand, color, speed

Behaviour: Drive, stop

3. Bank Account Class

A bank account can also be represented using a class.

```

class BankAccount {
    String accountHolder;
    double balance;

    void deposit(double amount) {
        balance = balance + amount;
    }

    void withdraw(double amount) {
        balance = balance - amount;
    }

    void displayBalance() {
        System.out.println("Balance: " + balance);
    }
}

public class Main {
    public static void main(String[] args) {

        BankAccount account = new BankAccount();

        account.accountHolder = "Rahul";
        account.balance = 5000;

        account.deposit(2000);
        account.withdraw(1000);

        account.displayBalance();
    }
}

```

Real-world entity: Bank Account

Properties: Account holder, balance
Behaviour: Deposit, withdraw, display balance

4. Employee Class

```
class Employee {  
    String name;  
    int employeeId;  
    double salary;  
  
    void work() {  
        System.out.println(name + " is working.");  
    }  
  
    void displayDetails() {  
        System.out.println("Name: " + name);  
        System.out.println("ID: " + employeeId);  
        System.out.println("Salary: " + salary);  
    }  
}
```

Real-world entity: Employee

Properties:

- Name
- Employee ID
- Salary

Behaviour:

- Work
 - Display details
-

5. Mobile Phone Class

```
class MobilePhone {  
    String brand;  
    String model;  
    int battery;  
  
    void call() {  
        System.out.println("Calling...");  
    }  
  
    void charge() {  
        battery = 100;  
    }  
}
```

```
        System.out.println("Phone is fully charged.");
    }
}
```

Real-world entity: Mobile Phone

Properties: Brand, model, battery

Behaviour: Call, charge

The easiest way to identify a class

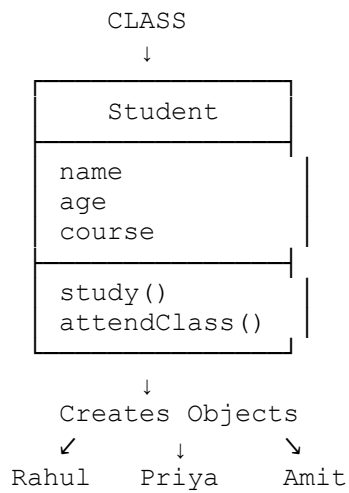
When looking at a real-world entity, ask two questions:

1. What does it HAVE? → Properties/Data

2. What can it DO? → Methods/Behaviour

Real-world Entity	Properties	Behaviour
Student	name, age, course	study(), attendClass()
Car	brand, color, speed	drive(), stop()
Bank Account	accountNo, holder, balance	deposit(), withdraw()
Employee	name, id, salary	work(), attendMeeting()
Mobile Phone	brand, model, battery	call(), charge()
Dog	name, breed, age	bark(), eat(), run()
Book	title, author, price	read(), displayDetails()
Laptop	brand, RAM, processor	start(), shutdown()

Important concept for students



So, in very simple words:

Class = Blueprint

Object = Real thing created from the blueprint

Properties = What the object has

Methods = What the object can do

For example, **car** is a class, while **myCar** is an object of the **car** class.